

In this chapter, we discuss a more advanced optimization algorithm using a case study. We'll consider the traveling salesman problem, as well as several of its possible solutions and their shortcomings. Finally, we'll introduce simulated annealing, an advanced optimization algorithm that overcomes these shortcomings and can perform global, rather than just local, optimization.

Life of a Salesman

The *traveling salesman problem (TSP)* is an extremely famous problem in computer science and combinatorics. Imagine that a traveling salesman wishes to visit a collection of many cities to peddle his wares. For any number of reasons—lost income opportunities, the cost of gas for his car, his head aching after a long journey ([Figure 6-1](#))—it's costly to travel between cities.